

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static · 35 trials
Training data path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static · 35 trials
Training data path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static · 35 trials
Training data path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static · 35 trials
Training data path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static · 35 trials
Training data path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static · 35 trials
Training data path 7	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static · 36 trials
Training data path 8	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static · 30 trials
Training data path 9	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static · 35 trials
Training data path 10	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static · 36 trials
Training data path 11	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static · 35 trials
Training data path 12	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static · 37 trials
Training data path 13	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static · 35 trials
Training data path 14	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Static · 0 trials
Training data path 15	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static · 35 trials
Training data path 16	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static · 35 trials
Total training paths	16 · 524 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static
Report output dir	/home/jestin/ThesisRepo/ML/NewReports/DynamicReports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	7: Double_Closed_Fist, Double_Nothing, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Feature mode	stats+fft → feature dim 9504
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=2095 · test=105

Augmentation (training only)

Enabled	yes
Copies per train trial	4
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.9, 1.1)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
Random Forest	1.0000	0.0000	0.9619
SVM (Linear)	1.0000	0.0000	0.9429
SVM (RBF)	1.0000	0.0000	0.9333
Logistic Regression	1.0000	0.0000	0.9333
LDA	1.0000	0.0000	0.7714
KNN	0.9933	0.0075	0.7619

Per-class hold-out F1

Class	Random Forest	SVM (Linear)	SVM (RBF)	Logistic Regression	LDA	KNN	Support
Double_Closed_Fist	1.000	0.968	0.968	1.000	0.812	0.774	16
Double_Nothing	0.846	0.875	0.828	0.828	0.583	0.667	14
Double_Okay	1.000	0.966	0.966	0.966	0.690	0.600	15
Double_Open_Palm	0.968	0.968	0.938	0.938	0.778	0.722	15
Double_Phone	0.933	0.933	0.929	0.966	0.714	0.692	15
Double_Pistol	0.968	0.889	0.933	0.889	0.897	0.909	15
Double_Spiderman	1.000	1.000	0.968	0.938	0.875	0.909	15
macro avg	0.959	0.943	0.933	0.932	0.764	0.753	105

External test (NewTestData)

Algorithm	External test acc
Logistic Regression	0.8857
Random Forest	0.8000
SVM (RBF)	0.7429
SVM (Linear)	0.7429
KNN	0.4000
LDA	0.4000

Per-class external accuracy

Class	Logistic Regression	Random Forest	SVM (RBF)	SVM (Linear)	KNN	LDA	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	0.400	1.000	5
Double_Nothing	1.000	0.800	1.000	0.800	0.400	0.400	5
Double_Okay	0.800	1.000	0.600	0.600	0.600	0.000	5
Double_Open_Palm	1.000	1.000	0.800	1.000	0.000	0.400	5
Double_Phone	1.000	0.800	0.800	0.800	0.200	0.000	5
Double_Pistol	0.400	0.000	0.000	0.000	0.400	0.200	5
Double_Spiderman	1.000	1.000	1.000	1.000	0.800	0.800	5